

SUE: Looking Before You Build

A visual diagnostic study of corporate sustainability report embeddings

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Corpus preview edition — Semiconductors subset

Abstract

We embed 3,685 chunks drawn from 13 publicly-available sustainability, ESG, and integrated reports (5 companies in the semiconductor equipment sector) into the 384-dimensional MiniLM sentence-embedding space and ask a diagnostic rather than a modeling question: what shape does this corpus have, and does that shape line up with properties an author would name for it? Our central observation is elementary and generalisable: when the content of a document class shifts, the mean embedding of that class shifts with it, and the displacement between class centroids is itself an interpretable signal. In the present corpus, the financial-disclosure passages of an integrated report and its narrative ESG passages sit at measurably distinct centroids, and a single linear direction separates them cleanly (3,447 vs. 238 chunks). The same direction accounts for a bimodal Gaussian-mixture fit ($\Delta\text{BIC} = -10,481$ in the top-20 principal-component subspace) that is invisible to any per-company or per-year partition. We argue that this centroid-shift-as-signal pattern is a general-purpose diagnostic for text collections whose author-declared structure ought to imprint on their embedding-space geometry, and we lay out its mathematical scaffolding (cosine, anisotropy, PCA, perplexity, BIC, Fisher’s ratio) before showing the corpus itself.

1 Introduction

1.1 What a Global Impact Report is

A Global Impact Report — variously titled “Sustainability Report,” “ESG Report,” “Corporate Responsibility Report,” or, when consolidated with the annual financial disclosure, “Integrated Report” — is an annual, predominantly non-financial disclosure document that a publicly-traded firm publishes for its investors, regulators, employees, customers, and civil society. Until roughly the middle of the last decade its publication was almost entirely voluntary; today its content is governed de facto by a growing patchwork of frameworks (GRI, SASB, TCFD, and the ISSB standards that have begun consolidating them) and, in some jurisdictions, de jure by regulation — most prominently the European Union’s Corporate Sustainability Reporting Directive (CSRD), which began phasing in mandatory disclosure obligations for large firms in 2024.

The document is a genre. Any given report will typically combine, in one PDF running 60 to 200 pages, several recurring content types: an opening letter from the chief executive framing the year’s priorities; a materiality assessment naming the topics the firm considers most consequential to its business and to its stakeholders; quantitative disclosure of greenhouse-gas emissions partitioned into Scope 1 (direct), Scope 2 (purchased energy), and Scope 3 (value-chain) categories; water withdrawal and consumption; waste and circularity metrics; workforce composition and safety statistics; supplier auditing and human-rights due-diligence procedures; governance disclosures covering board independence and committee structure; and a set of forward-looking targets, typically framed against a base year and a horizon (for example “a 50% absolute reduction in Scope 1 and 2 emissions from a 2019 baseline by 2030”).

Anatomy of a Global Impact Report — twelve recurring content types, coloured by rhetorical register

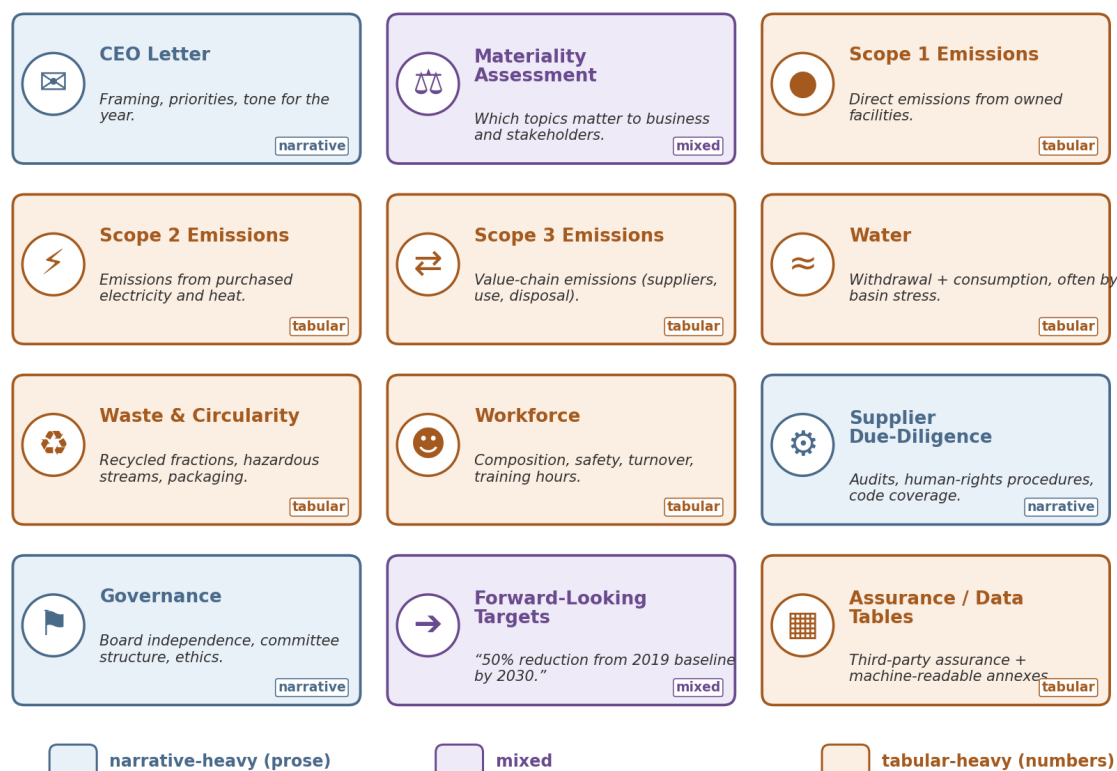


Diagram A: The twelve content types that recur across essentially every Global Impact Report in the corpus, coloured by rhetorical register. Narrative-heavy blocks (blue) read like corporate prose; tabular-heavy blocks (orange) are flattened spreadsheets of quantitative disclosure; mixed blocks (purple) combine both. The class-centroid axis of §2 is, empirically, the axis that separates the orange half from the blue half.

For a study of embedding-space geometry this genre is uniquely useful. Two rhetorical registers are placed inside the same document, on the same topics, at roughly the same length: narrative passages, which read like corporate prose, and tabular passages, which read like a spreadsheet flattened into text during PDF extraction. Both registers describe the same underlying commitments and the same underlying measurements. A sentence encoder that represents meaning ought to treat them as close; a sentence encoder that represents surface form ought to separate them. The corpus therefore admits a diagnostic question that most text collections do not: which of these two things is our encoder actually doing?

1.2 What is at stake in reading these documents well

Two questions animate the rest of the paper. The first is a disclosure question: for any given report, how many of its chunks land on the numeric side of the narrative↔tabular axis versus the narrative side, and what is each side actually saying? Are targets framed against comparable baselines, or is one firm quietly using a 2019 base year while another uses 2015? The class-centroid decomposition of §2 gives a first-cut, quantitative answer, and converts a stack of 3,000-page reports into an auditable quantity.

The second is a representation question: when an off-the-shelf sentence encoder is handed text of this genre — two rhetorical registers describing the same commitments on the same topics inside the same document — does it treat them as close (representing meaning) or does it separate them (representing surface form)? We will show empirically that it separates them: a single, learnable direction in embedding space cleanly divides prose

from tabular chunks. That direction is a discovered representational axis at the corpus level, of the same family as the axes mechanistic interpretability recovers at the token or circuit level.

Both questions turn out to be the same question in different vocabularies: what is inside these documents, and where is the boundary between what a system claims to represent and what it actually represents?

1.3 Why look at the vectors before building on top of them

Modern retrieval-augmented systems treat a corpus as a bag of vectors: each chunk is embedded, indexed, and retrieved by cosine similarity to a query embedding. The design of such a system rarely returns to look at the vectors themselves. This paper argues, through a small, replicable case study, that the vectors are worth looking at: naive assumptions about corpus homogeneity fail on data that would seem, on paper, to be as homogeneous as it gets.

Our corpus is a set of Global Impact, Environmental Social and Governance, and Integrated Reports published by the five largest semiconductor-equipment firms: ASML, Applied Materials, KLA, Lam Research, TEL. Every document is written for a comparable audience — investors, regulators, employees, journalists — on a comparable set of topics — emissions, water, supply chain, workforce, governance. The naive prediction is that the sentence embeddings of chunks drawn from these documents form a single blob: a lot of overlap, little separation. That prediction is wrong, and its being wrong is instructive.

The paper is organised around a single claim and its consequences. §2 states the claim in its most compact form: for a text collection whose authors mix identifiable content classes into one document, the class centroids in embedding space are displaced from one another, and that displacement is a rich, interpretable signal. §3 is a mathematical warm-up — a short primer on the objects (cosine similarity, variance, anisotropy, perplexity, BIC, Fisher’s ratio) that the rest of the paper is written in terms of. §4 introduces the corpus itself. §5 exhibits the first-order surprise the abstract advertised: the point cloud is bimodal, and the split is not by company or year. §6 quantifies anisotropy and clarifies what PCA does and does not see. §7 replays the corpus under nonlinear projections (t-SNE, UMAP). §8 discusses what changes downstream once the centroid structure is admitted, and §9 sketches the research directions this study opens — including where else the centroid-shift diagnostic is likely to be productive.

1.4 How to read this paper

This paper is written with a reader in mind who is comfortable reading, but not necessarily fluent in, the mathematics of high-dimensional statistics. Every equation is introduced by an intuition first, unpacked symbol by symbol second, and used in a figure third. If a piece of notation feels new — the covariance matrix, Bayesian information criterion, perplexity, Fisher’s discriminant ratio — it is defined in §3 before it is used later. Readers already fluent in this material can skim or skip §3 without missing the argument.

Every empirical claim in the paper is accompanied by either a figure or a quotation of a real chunk from the corpus, so that a reader can audit the claim rather than take it on trust. The corpus itself, the embedding matrix, and the code that produces every figure and number in this document are included in the accompanying repository.

2 Centroids as a data-story engine

The technical measurements in the remainder of this paper all serve a single, elementary claim. State it first so the figures can be read as evidence for it, rather than as a sequence of unmotivated diagnostics.

Claim. Given a text corpus C and a partition of its chunks into content-defined classes $C = C_1 \sqcup \dots \sqcup C_k$, and any well-behaved sentence encoder E , the class centroids

$$\mu_j = \frac{1}{|C_j|} \sum_{x \in C_j} E(x) \in \mathbb{R}^{384} \quad (1)$$

are displaced from one another whenever the classes are genuinely different in content, and the displacements $\mu_i - \mu_j$ carry interpretable information about how the classes differ. Shifted content $\rightarrow$ shifted embeddings $\rightarrow$ shifted centroid. The equation is a bookkeeping identity; the empirical claim is that in real corpora the displacement is measurable, stable, and human-readable.

This is not a subtle statement. Its power is that the converse chain runs in the useful direction: a measurable centroid displacement in embedding space is evidence of a content distinction that a partition may or may not already know about. If the classes are known in advance, the displacement quantifies their separability. If the classes are not known, unsupervised clustering that recovers the same displacement retrieves the latent content axis. The rest of this paper does both, on one worked corpus.

2.1 The minimal example: financial disclosure vs. narrative ESG

The corpus that motivates the paper is a set of integrated, sustainability, and ESG reports (§4) that mixes, within the same document, two content classes an author would readily name: financial-disclosure passages — income statements, balance-sheet fragments, line-item tables of costs, revenues, and emissions figures — and narrative ESG passages — prose about strategy, materiality, and progress against targets. Both classes describe overlapping subject matter, in the same document, at comparable length. Only the register differs. We assign each chunk to one class by a light regex-derived label (digit density, line-length statistics), which yields 3,447 narrative chunks and 238 tabular chunks.

Linear discriminant analysis on this two-class label recovers a single direction $w \in \mathbb{R}^{384}$ along which the two class centroids are maximally separated relative to their internal spread. Figure 1 projects the corpus onto that one-dimensional axis; the two class distributions are cleanly separated, and their means — the projections of $\mu_{\text{narrative}}$ and μ_{tabular} — sit far apart on a scale where within-class spread is small. Content shifts; centroids shift; the shift is visible in one dimension.

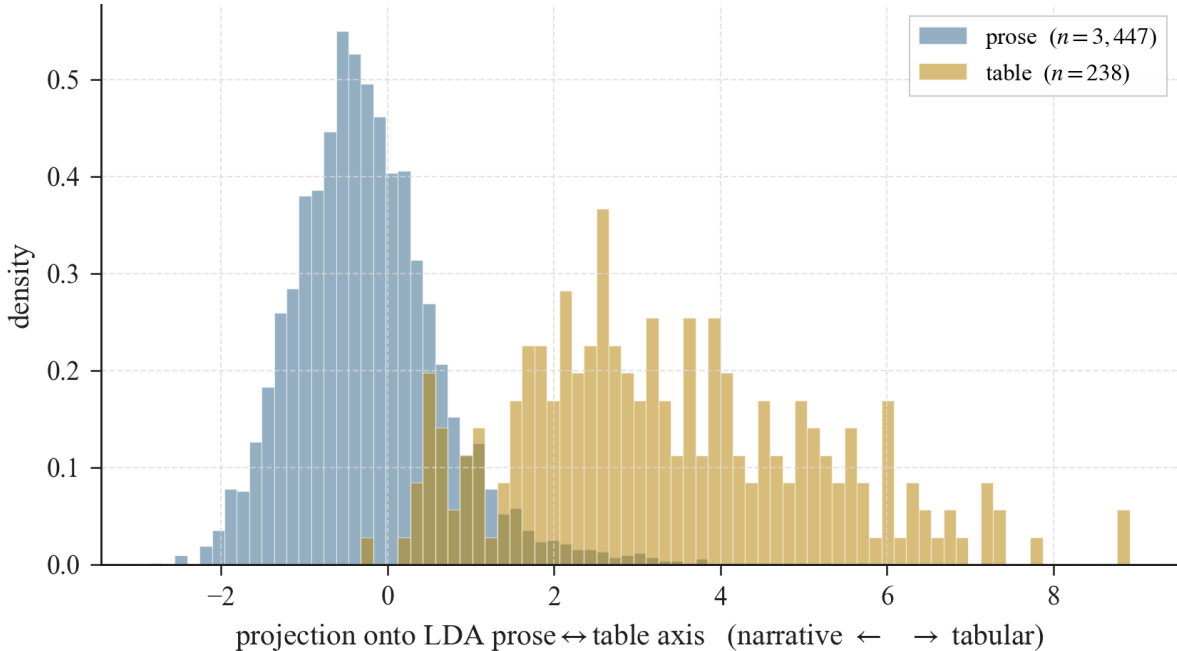


Figure 1: Projection of every corpus chunk onto the single direction that maximises the ratio of between-class to within-class scatter (Fisher’s ratio, defined in §3.7). The two class centroids — narrative and tabular — are displaced along this axis by many within-class standard deviations. This is the paper’s central image.

2.2 Why the centroid displacement is the whole story

Two independent lines of evidence, developed in later sections, converge on the same axis. First, an unsupervised Gaussian-mixture fit to the raw embeddings (§5) discovers two components without being told about the regex labels. Second, when the corpus is projected onto the supervised centroid axis of Figure 1 and coloured by the unsupervised cluster assignment (Figure 2 below), the two clusterings coincide almost exactly. The label the corpus would give itself is the label the geometry has already found.

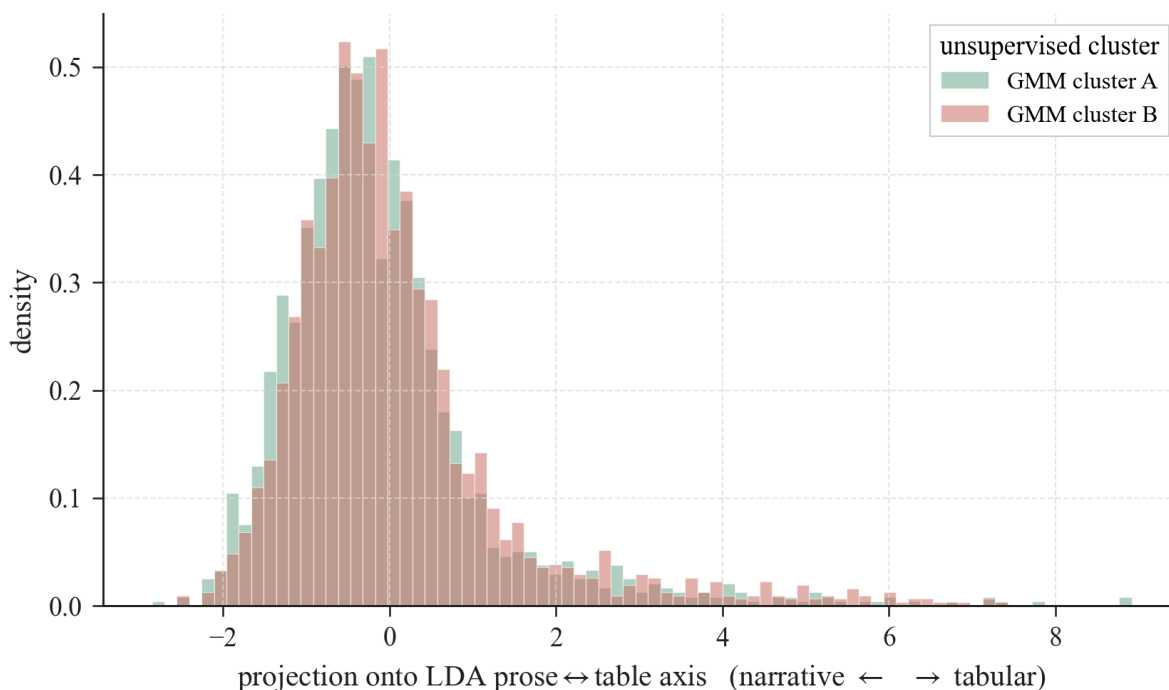


Figure 2: Chunks re-projected onto the supervised centroid axis of Figure 1, coloured by the unsupervised two-component Gaussian mixture fit of §5. The two assignment schemes agree: the bimodality of the raw embedding cloud lives on the class-centroid displacement.

2.3 Why this generalises

Nothing in the claim is specific to sustainability disclosure. The same construction applies to any corpus whose author-declared structure mixes identifiable content classes into a single stream of text: scientific papers (methods vs. discussion), medical records (structured history vs. free-text note), legal filings (statutory citation vs. argument), transcripts (speaker A vs. speaker B), or the technical vs. rhetorical passages of a research grant. In each case the centroid displacement $\mu_i - \mu_j$ is a one-line diagnostic that can be computed, plotted, and audited before any retrieval or downstream model is built on top of the embeddings. Where a domain lacks an author-supplied label, an unsupervised mixture fit followed by centroid inspection recovers the same information; we execute both routes in §5. We return to further application domains in §9.

The rest of the paper is scaffolding for this claim: §3 defines the mathematical objects, §4 introduces the corpus, §5–§7 measure the geometry that makes the claim rigorous, §8 discusses downstream implications, and §9 catalogues where the diagnostic should be pushed next.

3 Mathematical warm-up

This section defines every object the rest of the paper uses, in plain language and with a picture in mind before any algebra. A reader already fluent in embeddings, PCA, entropy, and BIC can skim; a reader for whom any of the following objects is new should read this section carefully — every subsequent figure is stated in these

terms.

3.1 What an embedding actually is

An embedding is a function that maps a piece of text — a word, a sentence, or in our case a ~900-character chunk — to a point in a high-dimensional space. For us that space is $\mathbb{R}^{384}$: every chunk becomes a list of 384 real numbers. Formally, if T is the set of possible text inputs, an embedding is a map

$$E : T \rightarrow \mathbb{R}^{384},$$

and the numbers themselves are meaningless in isolation; what matters is relative geometry. If two chunks land at nearby points, the encoder is claiming they mean similar things. If they land far apart, the encoder is claiming they mean different things. The goal of training an encoder is precisely to arrange the mapping so that this geometric claim is reliable.

The encoder we use, all-MiniLM-L6-v2, additionally unit-normalises its outputs: after producing $x \in \mathbb{R}^{384}$, it divides by the length $\|x\|$ so that every embedded chunk lies on the surface of the unit sphere $S^{383} \subset \mathbb{R}^{384}$. This means every geometric statement we will make can be made either as a statement about angles or as a statement about distances — the two are the same object on the sphere.

3.2 Cosine similarity, three ways

The cosine similarity of two vectors is the cosine of the angle between them, viewed as arrows from the origin. Three equivalent expressions are worth having in mind:

$$\cos(a, b) = \frac{\langle a, b \rangle}{\|a\| \|b\|} = \langle \hat{a}, \hat{b} \rangle = 1 - \frac{1}{2} \|\hat{a} - \hat{b}\|^2,$$

where $\langle a, b \rangle$ is the dot product $\sum_i a_i b_i$, and $\hat{a}, \hat{b}$ are the unit-normalised versions of a, b . The three forms say the same thing: cosine similarity is the dot product of the normalised vectors, and on the unit sphere, cosine and Euclidean distance are locked together. Its range is $[-1, 1]$: 1 when the two vectors point the same way, 0 when they are perpendicular, and -1 when they point opposite. For random unit vectors in a well-behaved (isotropic) space, the expected value is 0. We will use that fact as a diagnostic in §6.1.

3.3 Variance, covariance, and principal components

The variance of a random variable measures how spread out its values are, on average, around its mean. In one dimension:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

Zero variance means every sample is exactly the mean; large variance means samples are far-flung. The two-variable generalisation is covariance:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)],$$

which is positive when X and Y tend to rise and fall together, negative when they tend to move in opposite directions, and zero when they are linearly unrelated. Stacking these covariances into a matrix for a 384-dimensional vector gives the covariance matrix $\Sigma \in \mathbb{R}^{384 \times 384}$, whose entry Σ_{ij} is $\text{Cov}(X_i, X_j)$.

Principal component analysis (PCA) diagonalises Σ . Its eigenvectors are called principal components; each eigenvalue tells us how much of the total variance lies along its eigenvector. PC1 is the direction along which the corpus is most spread out, PC2 is the next direction orthogonal to it along which it is next-most spread out, and so on. Projecting a 384-dimensional cloud onto its first two principal components gives the flat picture that every subsequent scatter plot in this paper is drawn on. One caveat is worth flagging now, and returning to in §6.2: PCA subtracts the mean before it looks for variance. Any structure that lives in the mean direction of the

corpus is, by construction, invisible to PCA.

3.4 Anisotropy: when the cloud has a preferred direction

A cloud of points is isotropic if it has no preferred direction — a spherical swarm of gnats. It is anisotropic if it does — a school of fish swimming north. Formally, a corpus of unit vectors is isotropic when

$$\mathbb{E}_{i \neq j} [\cos(x_i, x_j)] \approx 0,$$

and anisotropic when this expectation is strictly positive. Sentence encoders trained with self-supervised objectives are known to be strongly anisotropic (Ethayarajh, 2019): their outputs occupy a narrow cone rather than filling the sphere. Practically, this means the “baseline” cosine similarity between two unrelated chunks is not 0 but something distinctly positive, and every retrieval decision that a downstream system makes is being made against that offset. We measure the offset directly in §6.1.

3.5 Entropy, perplexity, and effective neighbourhood size

Perplexity is a knob in t-SNE (§7) whose meaning is not obvious from its name. It comes from information theory, and it is easiest to explain by first defining entropy.

The Shannon entropy of a probability distribution $P = (p_1, \dots, p_k)$ over k outcomes is

$$H(P) = - \sum_{i=1}^k p_i \log_2 p_i.$$

It is measured in bits, and the intuition is: on average, how many yes/no questions do I need to ask to identify a sample from this distribution? A uniform distribution over 8 outcomes has entropy $\log_2 8 = 3$ bits (three yes/no questions is exactly what a binary search needs). A distribution concentrated on a single outcome has entropy 0. Entropy is a measure of how uncertain, or how spread out, a distribution is.

Perplexity is simply the exponential of entropy:

$$\text{Perp}(P) = 2^{H(P)}.$$

It converts “bits of uncertainty” into an effective number of choices. A fair coin has entropy 1 bit and perplexity 2 (two equally-plausible outcomes). A 99/1 biased coin has entropy near 0 and perplexity near 1 (effectively one outcome). A uniform distribution over 30 outcomes has perplexity 30. Perplexity is entropy in a unit the eye is trained to read.

In t-SNE, we do the following: for each point x_i , we place a Gaussian “attention” distribution over all other points, with the Gaussian’s bandwidth σ_i tuned so that the resulting distribution has some chosen perplexity. Because perplexity is “effective number of neighbours,” setting perplexity = 30 is telling the algorithm each point should care about roughly 30 nearest neighbours when deciding where to land in the 2-D map. Small perplexity → hyper-local emphasis, many small clumps. Large perplexity → global emphasis, one big shape. §7 renders the same corpus at four perplexity choices side by side.

One more information-theoretic object is worth introducing briefly, because t-SNE and its cousins optimise it: the Kullback–Leibler divergence from a distribution Q to a distribution P ,

$$D_{\text{KL}}(P \parallel Q) = \sum_i p_i \log \frac{p_i}{q_i},$$

measures how surprised you would be, on average, if you believed the world was Q and it turned out to be P . It is zero when $P = Q$ and grows otherwise. t-SNE builds a neighbourhood distribution P in the original 384-dimensional space and a neighbourhood distribution Q in the 2-D map, and arranges the 2-D coordinates

so that $D_{\text{KL}}(P \parallel Q)$ is as small as possible.

3.6 The Bayesian Information Criterion

How do we decide whether a two-cluster model of the corpus is better than a one-cluster model? Any two-cluster model has strictly more free parameters, so it will fit the data strictly better in a raw likelihood sense. The Bayesian Information Criterion (BIC) trades fit against complexity:

$$\text{BIC}(k) = k \ln N - 2 \ln \hat{L}_k,$$

where k is the number of free parameters, N the number of datapoints, and $\hat{L}_k$ the maximised likelihood of the model. Lower BIC is better. Adding a parameter helps only if it improves the log-likelihood by at least $\ln N \div 2 \approx 4.1$ for our $N = 3,685$. This is a soft Ockham’s razor with a specific numerical threshold. In §5 we use $\Delta\text{BIC} = \text{BIC}_2 - \text{BIC}_1$ to ask whether the corpus is better described by two Gaussian clusters than by one.

3.7 Fisher’s discriminant ratio

Given points labelled with one of two classes — in our case, chunks tagged as “prose” or “table” by a regex — linear discriminant analysis (LDA) finds the direction w along which the two classes are maximally separated relative to their own internal spread. It maximises

$$J(w) = \frac{w^\top S_B w}{w^\top S_W w},$$

where S_B is the between-class scatter (how far the class means are from the overall mean, weighted by class size) and S_W is the within-class scatter (how spread out each class is around its own mean). The ratio is a signal-to-noise ratio: it grows when class means separate and shrinks when either class becomes more diffuse. Fisher’s ratio is the object underlying Figure 1 of §2: the direction it selects is what makes the centroid displacement legible in one dimension.

3.8 A note on empirical sampling

Several of the distributions in the rest of the paper — the cosine-similarity histograms and the anisotropy reference of §6 — are constructed by sampling pairs rather than enumerating all of them. With $N = 3,685$ chunks the number of pairs is $N(N-1)/2 \approx 6.8 \times 10^6$, which is computationally awkward and, more importantly, unnecessary: a random sample of 20,000 to 50,000 pairs gives a Monte-Carlo estimate of any distributional summary whose standard error is $O(1/\sqrt{K})$, i.e. below the third decimal place. Every empirical distribution reported below is such a sample; every seed is fixed for reproducibility.

4 The corpus

After light PDF text extraction, cleaning, and fixed-size chunking (approximately 900 characters, 120-character overlap), the corpus contains 3,685 chunks across 13 reports; chunk counts per company (Figure 3) vary from about 250 to about 950 per firm, an imbalance inherited from publishing conventions rather than curation. Each chunk is passed through *all-MiniLM-L6-v2*, producing a 384-dimensional unit-normalised embedding. On unit vectors, cosine similarity and Euclidean distance are monotonically related through $\|a - b\|^2 = 2(1 - \cos(a, b))$, so every statement about distance below can be read equivalently as a statement about cosine similarity.

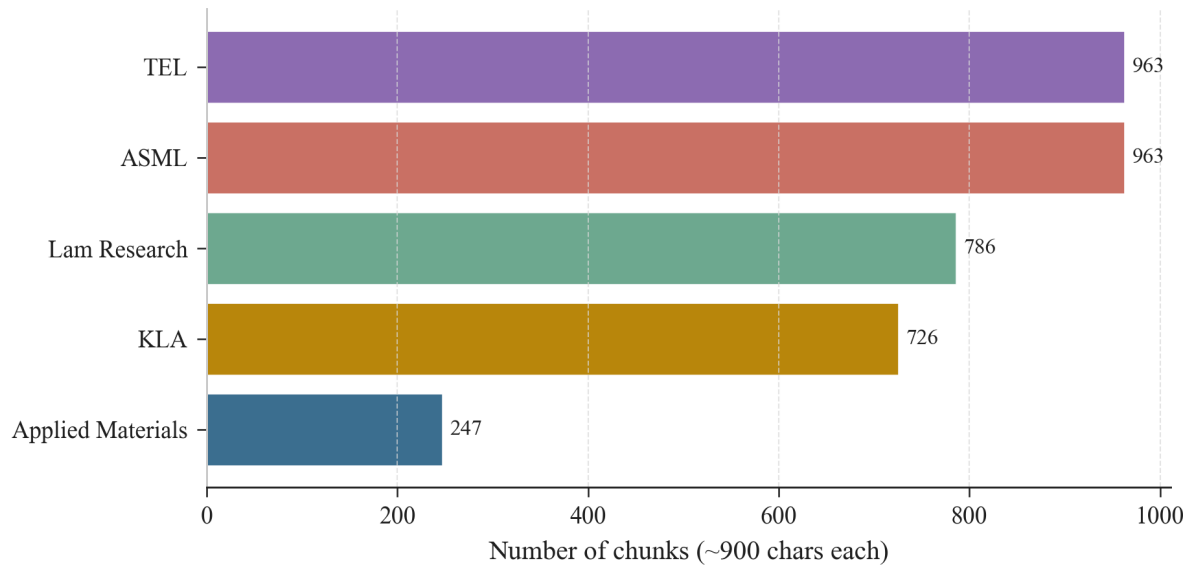


Figure 3: Chunks per company. Every chunk is a contiguous ~900-character window of extracted PDF text.

5 Bimodality: the corpus splits, but not by author

Figure 4 shows a principal-component projection of the whole corpus to two dimensions, coloured by publishing company. Two features are immediate. First, the colours mix: the five companies do not occupy distinguishable regions. Any per-firm retrieval design that assumed firm-level clustering would be building on sand. Second, the cloud is not one lump; there are two visible modes.

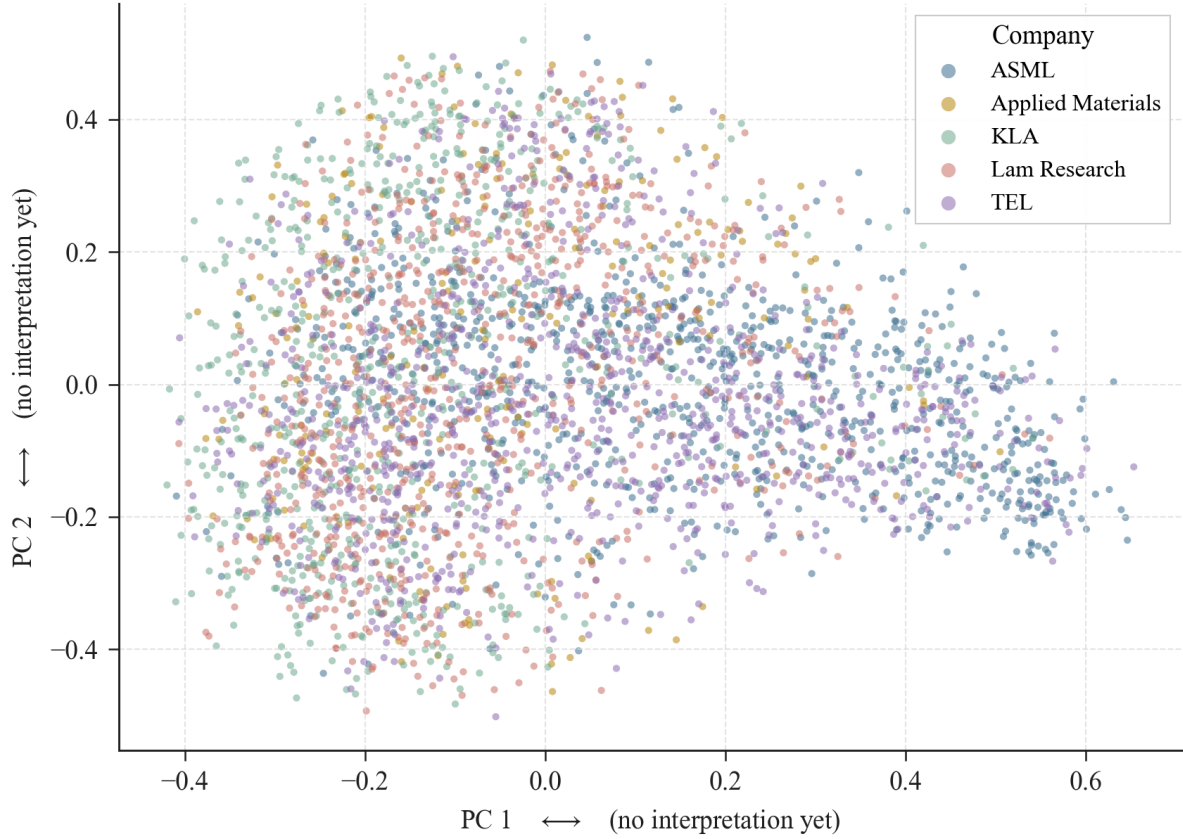


Figure 4: Two-dimensional principal-component projection of the 3,685 corpus chunks, coloured by publishing company. The colours mix; the shape does not. Company identity is not the axis along which this space is structured.

To promote the visual impression to a measured claim we fit two Gaussian mixtures — with one and two full-covariance components respectively — and compare their BIC. A methodological subtlety is worth flagging: in the raw 384-dimensional space each additional full-covariance component introduces roughly $384 + C(384, 2) \approx 74,000$ free parameters, and the BIC penalty $k \ln N$ swamps any log-likelihood improvement even when a two-cluster fit is visually obvious. We therefore fit both mixtures in the top-20 principal-component subspace, which retains 50% of corpus variance and reduces the per-component parameter count to 230. In that subspace $\Delta\text{BIC} = -10,481$, a decisive preference for two clusters (Figure 5).

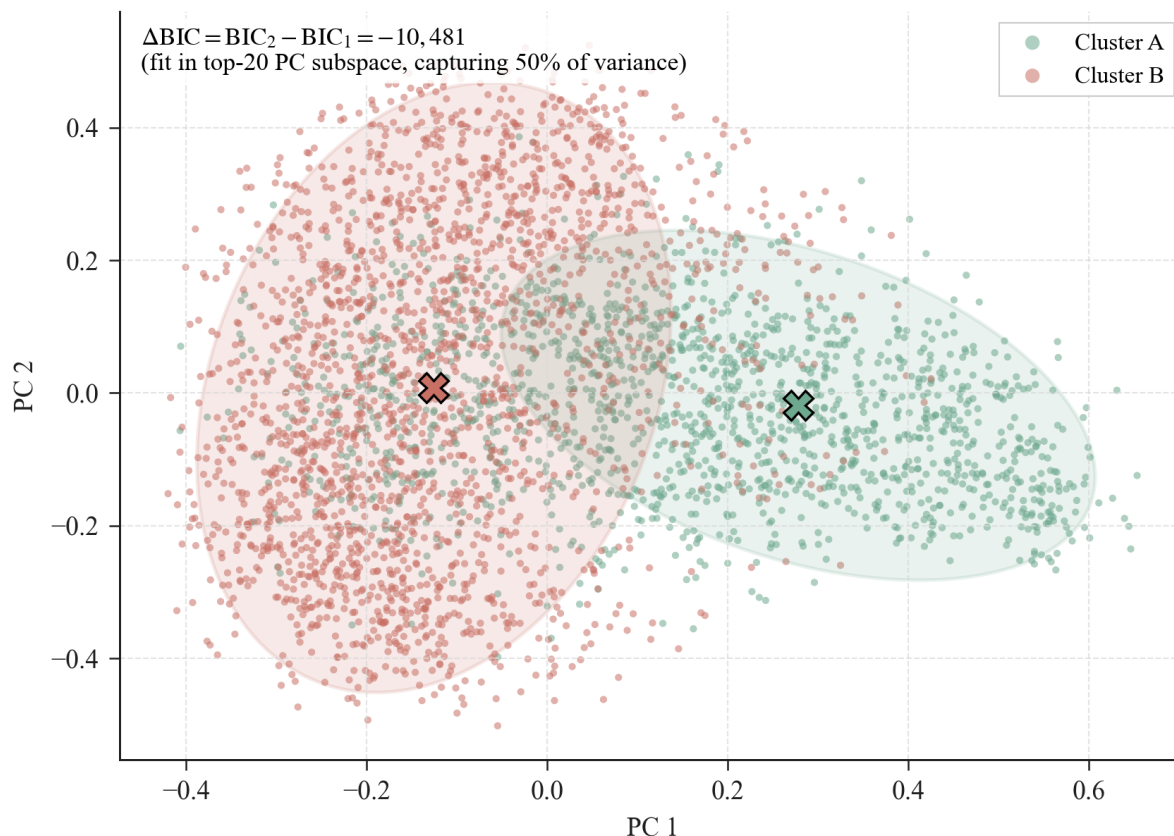


Figure 5: A two-component Gaussian mixture fitted in the top-20 principal-component subspace, projected here onto the same principal-component plane as Figure 4. Ellipses are two standard deviations of each cluster's projected covariance. The bimodality is a property of the embedded corpus, not of the two-dimensional view.

A representative chunk from one cluster reads:

“DG, a not-for-profit, public-private partnership. In the semiconductor industry, in which the speed of technological innovation is rapid, developing new technologies in advance is a source of corporate growth. We will not only develop leading-edge exposure technology but cont”

— TEL, Tokyo Electron Limited (TEL) 2024 Integrated Report (SustainabilityReports.com).pdf, chunk 128 (one cluster)

and a representative chunk from the other reads:

“1.0) (13.7) (3,271.4) (11.9) 13.2 Total cost of sales (10,473.3) (49.5) (13,422.4) (48.7) 28.2 Gross profit 10,700.1 50.5 14,136.1 51.3 32.1 Research and development costs (3,253.5) (15.4) (3,980.6) (14.4) 22.3 Selling, general and administrative costs (945.9) (4.5) (1,113.2)”

— ASML, 2023 ASML Annual Report.pdf, chunk 181 (the other cluster)

The first is narrative ESG prose. The second is what PDF-to-text extraction produces when it walks over a table of financial line items: numbers, parenthesised signs, whitespace where columns used to be, no verbs. These are the two class centroids of §2 made concrete: the unsupervised mixture has discovered the same partition that a regex-based content labeller produces, and the class means displaced along Fisher's direction are what the eye is seeing in Figures 4-5.

6 Reading cosine similarity honestly

Cosine similarity is the atom of every retrieval decision that follows: to ask “which chunk is most like this query?” is to sort the corpus by it. It is therefore worth knowing what its distribution looks like across pair types we would expect to differ. Figure 6 partitions the corpus into four pair-type populations. Same-document pairs average $\bar{x} \approx 0.34$; same-company pairs average 0.34; different-company pairs average 0.29. Two observations. First, the same-document and same-company distributions are essentially coincident — multiple annual reports from the same firm on the same topics are as textually related as chunks within a single report. Second, and more importantly, none of these distributions is centred near zero. Even different-company pairs cluster near 0.29. That offset is a property of the encoder, not of our corpus.

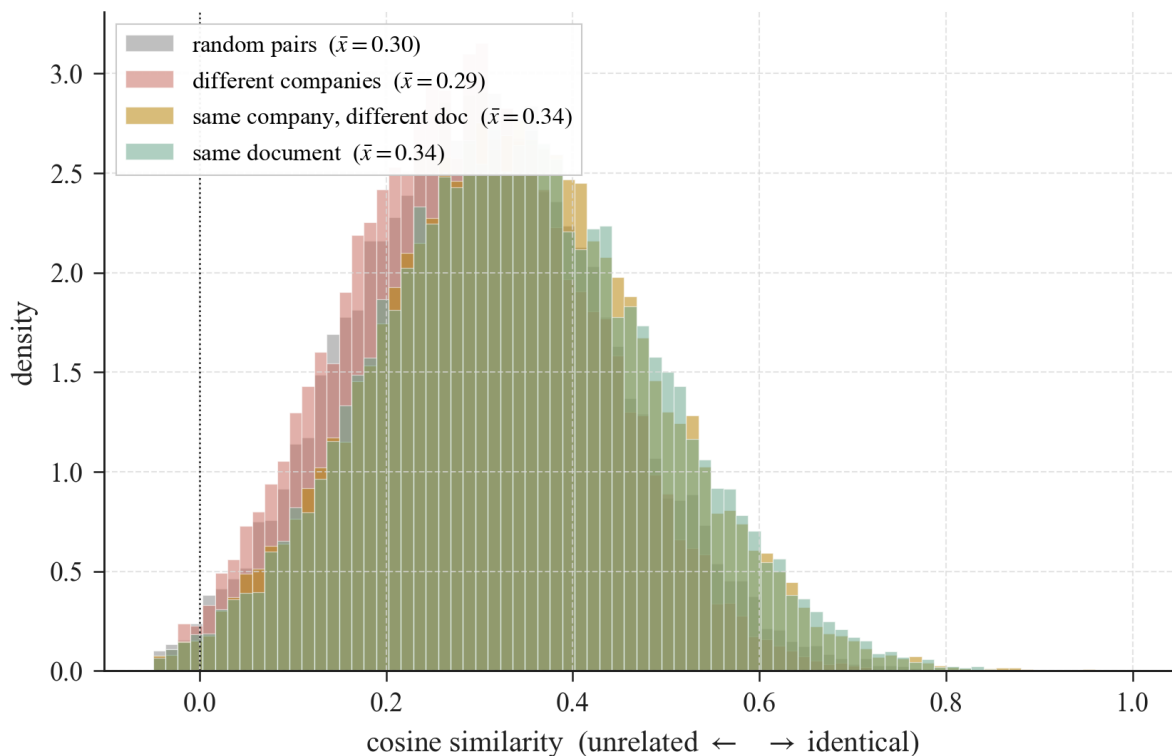


Figure 6: Pairwise cosine-similarity distributions across four pair populations. Within-document and within-company distributions coincide almost exactly. All four sit well above zero — the subject of §6.1.

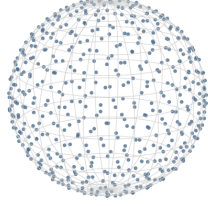
How to read Figure 6. Each coloured histogram is the answer to one question: if I pick two chunks from a particular relationship — same document, same company, different companies, or random pairs — how similar do they typically look to MiniLM? The x-axis is cosine similarity: 0 means “unrelated,” 1 means “identical.” The story the plot tells is a double surprise. First, the four histograms nearly coincide: MiniLM barely distinguishes two random reports from two chunks of the same report. Second, none of them is centred at zero. Even random-pair similarity averages 0.29, which is where the next subsection starts.

6.1 Anisotropy

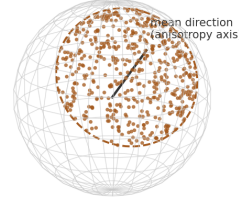
An isotropic embedding space would satisfy $E_{i \neq j}[\cos(x_i, x_j)] \approx 0$ for random chunk pairs. In practice pretrained contextual encoders exhibit strong anisotropy (Ethayarajh, 2019): outputs occupy a narrow cone, and the left-hand side lands well above zero. On our corpus $\bar{c}_{\text{SUE}} = 0.299$ against an isotropic reference $\bar{c}_{\text{iso}} = 0.000$ (Figure 7). The gap between the means is the quantity Ethayarajh names, and every cosine we report should be read against this offset.

Why every cosine in the SUE corpus sits above zero: anisotropy is a first-moment property of the encoder

Isotropic space (well-behaved)



Anisotropic space (what MiniLM does)



Every direction is used equally; two random points are typically far apart on the sphere. In anisotropic space, embeddings crowd into a cone; even ‘unrelated’ pairs are close together.

mean $\cos(\text{random pair}) \approx 0.00$

mean $\cos(\text{random pair}) \approx +0.30$ (real SUE value)

Read the cosine-similarity histograms as measurements against the right-hand cone, not the left-hand sphere. The “offset” in every distribution is the encoder’s cone half-angle made numeric.

Diagram B: Anisotropy, without algebra. Left: an isotropic embedding space; every direction is used equally and the mean cosine between random pairs sits at 0. Right: an anisotropic space, of the kind MiniLM produces; all embeddings crowd into a cone around a preferred axis, so even “unrelated” pairs land at cosine $\approx +0.30$. The cone half-angle is exaggerated for legibility; the quoted $+0.30$ is the measured SUE value. Every histogram in Figure 6 is a slice through the right-hand cone.

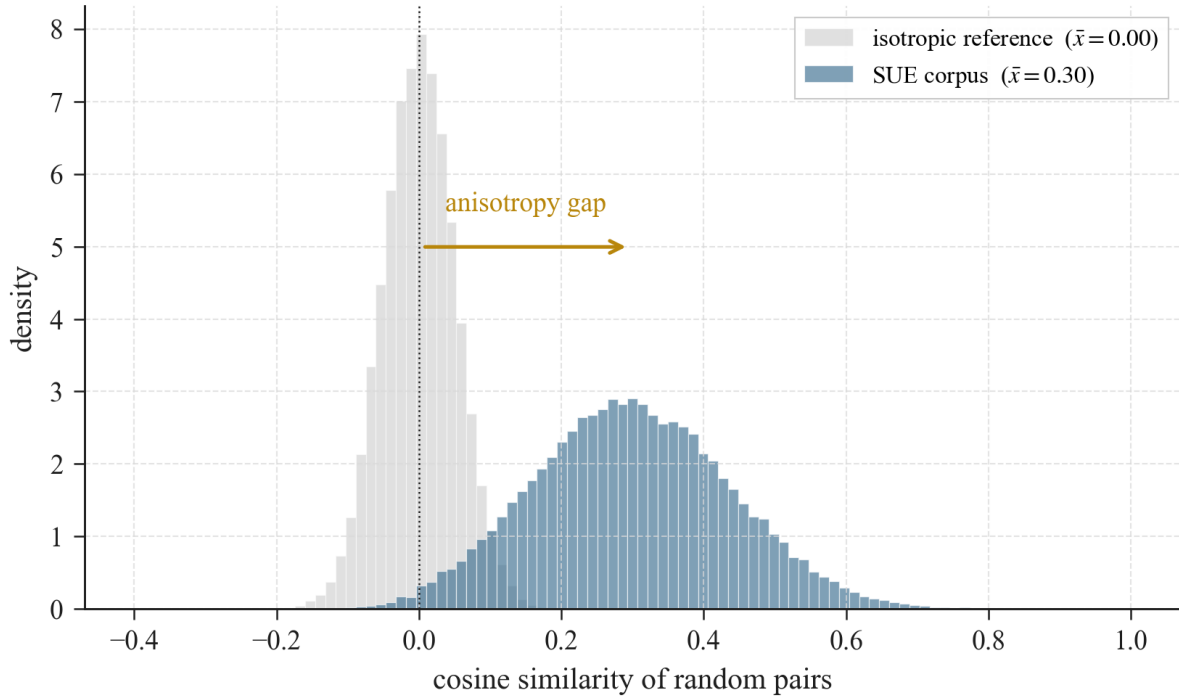


Figure 7: Cosine similarity between random pairs in our corpus (blue) against an isotropic reference (grey). The gap between the means is a property of the encoder, not of the corpus.

How to read Figure 7. The grey histogram is what cosine similarity would look like if MiniLM lived on the full sphere of Diagram B (mean 0). The blue histogram is what it actually looks like on our corpus (mean 0.30). The horizontal gap between the two means is the anisotropy of the encoder made numeric: it is the cosine of the cone’s half-angle, expressed as a similarity. Once we subtract that offset, the “signal” that same-document pairs

carry above random pairs is small — roughly 0.05 in Figure 6 — and every retrieval decision the encoder makes is resolving that thin margin.

6.2 What PCA does and does not see

A common intuition — worth explicitly dismantling — is that PC1 of an anisotropic corpus is dominated by the anisotropy axis. It is not. Anisotropy is a first-moment property: the mean direction $\bar{x}$ is far from zero. PCA operates on the centred data matrix: it subtracts the mean before diagonalising the covariance. The direction along which anisotropy concentrates the mean is therefore invisible to PCA. What Figure 8 shows is the residual, second-moment structure of the corpus after that offset has been removed.

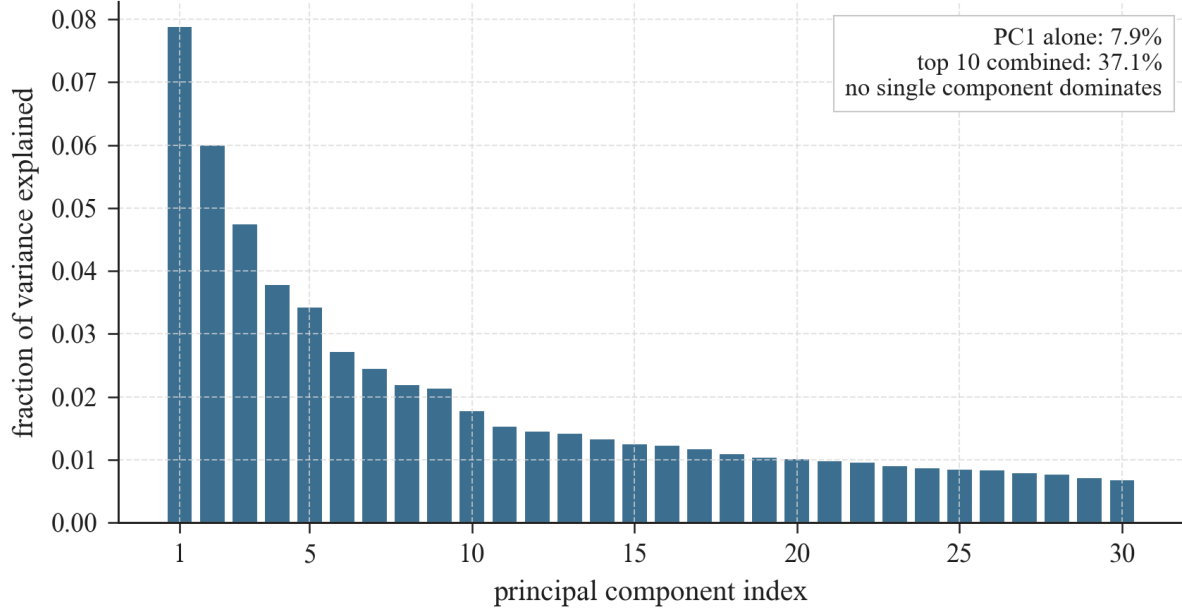


Figure 8: Fraction of variance explained by each of the first thirty principal components. PC1 accounts for 7.9% of the total variance, the first ten combined for 37.1%. No single component dominates.

On our corpus PC1 accounts for 7.9% of variance and the first ten combined for 37.1%: a relatively flat spectrum, meaning the two-dimensional principal-component projections of Figures 4-5 are picking up genuine, if partial, second-moment structure — they are not renderings of the anisotropy vector. Whitening still has a purpose. A linear approximation to Li et al. (2020) is per-dimension mean/variance normalisation $\tilde{x} = \text{diag}(\sigma)^{-1}(x - \mu)$, after which the covariance is approximately the identity and the mean is at the origin by construction. Figure 9 compares the corpus before and after this transformation, each projected to its own top-two principal components. The two-lobe shape persists — the important observation. It was not an artefact of anisotropy or of the mean offset.

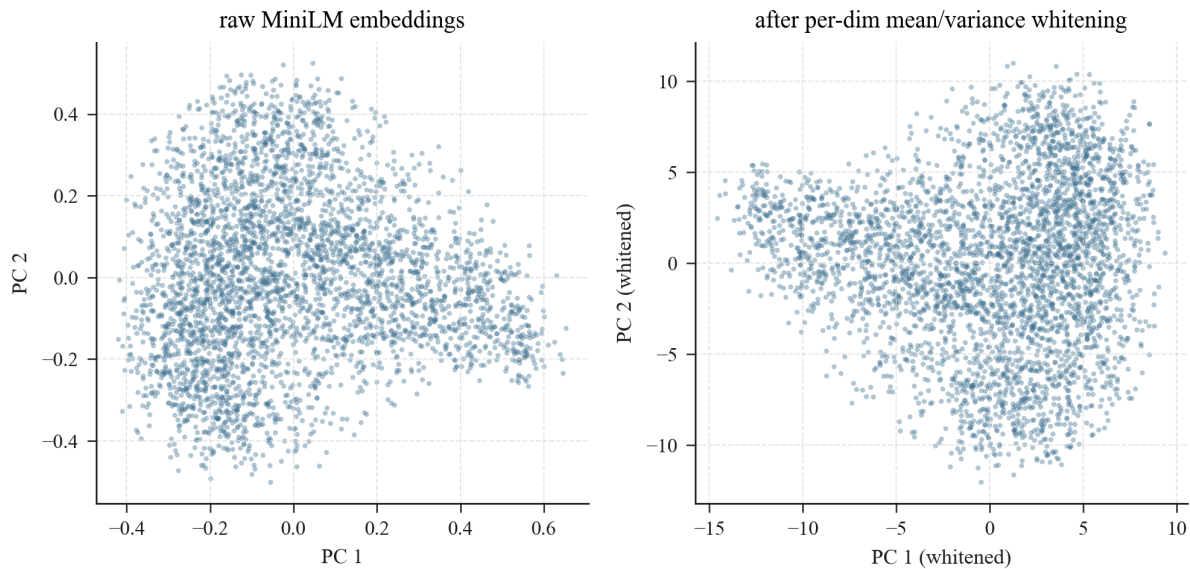


Figure 9: Left: raw MiniLM embeddings under two-dimensional principal-component projection. Right: after per-dimension mean/variance whitening, projected to its own top-two principal components. The tilt flattens; the two-lobe structure persists.

7 Nonlinear projections: t-SNE and UMAP

Nothing forces us to use principal components. t-SNE (van der Maaten and Hinton, 2008) and UMAP (McInnes et al., 2018) are the standard nonlinear alternatives; both have visible hyperparameters that materially change the picture. Figures 10 and 11 replay the corpus under four settings each. The purpose is not to nominate a winner; it is to make the reader aware that every low-dimensional plot of a corpus is a rendering through a chosen lens. Across all settings, the two-lobe structure of §2 and §5 is preserved — the class-centroid displacement is robust to the choice of projection, which is the essential invariance for our claim.

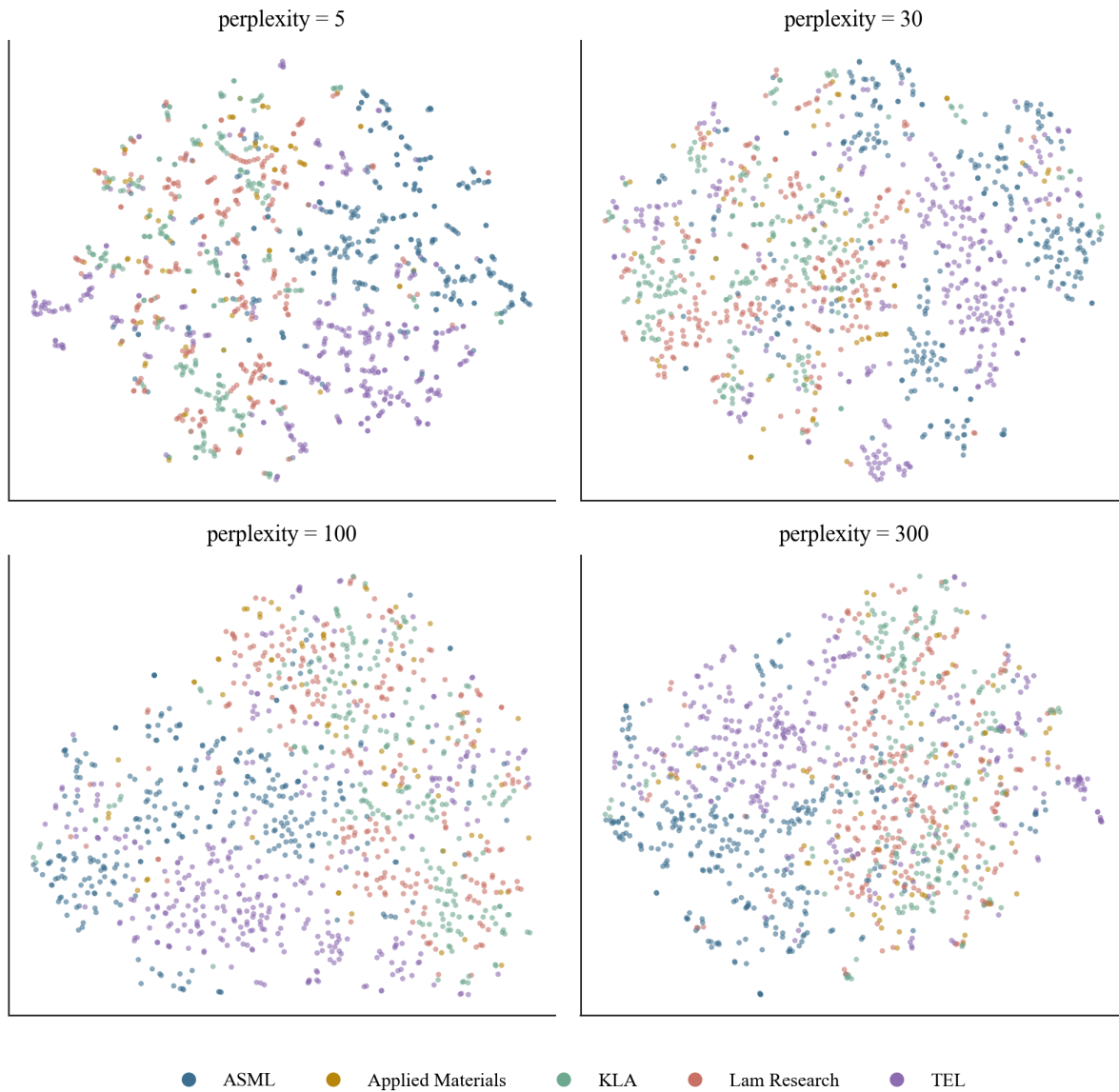


Figure 10: *t-SNE projections of a 1200-chunk stratified sample at four perplexity settings. Same data, same seed, four different neighbourhood-scale choices; the two-lobe structure survives every setting.*

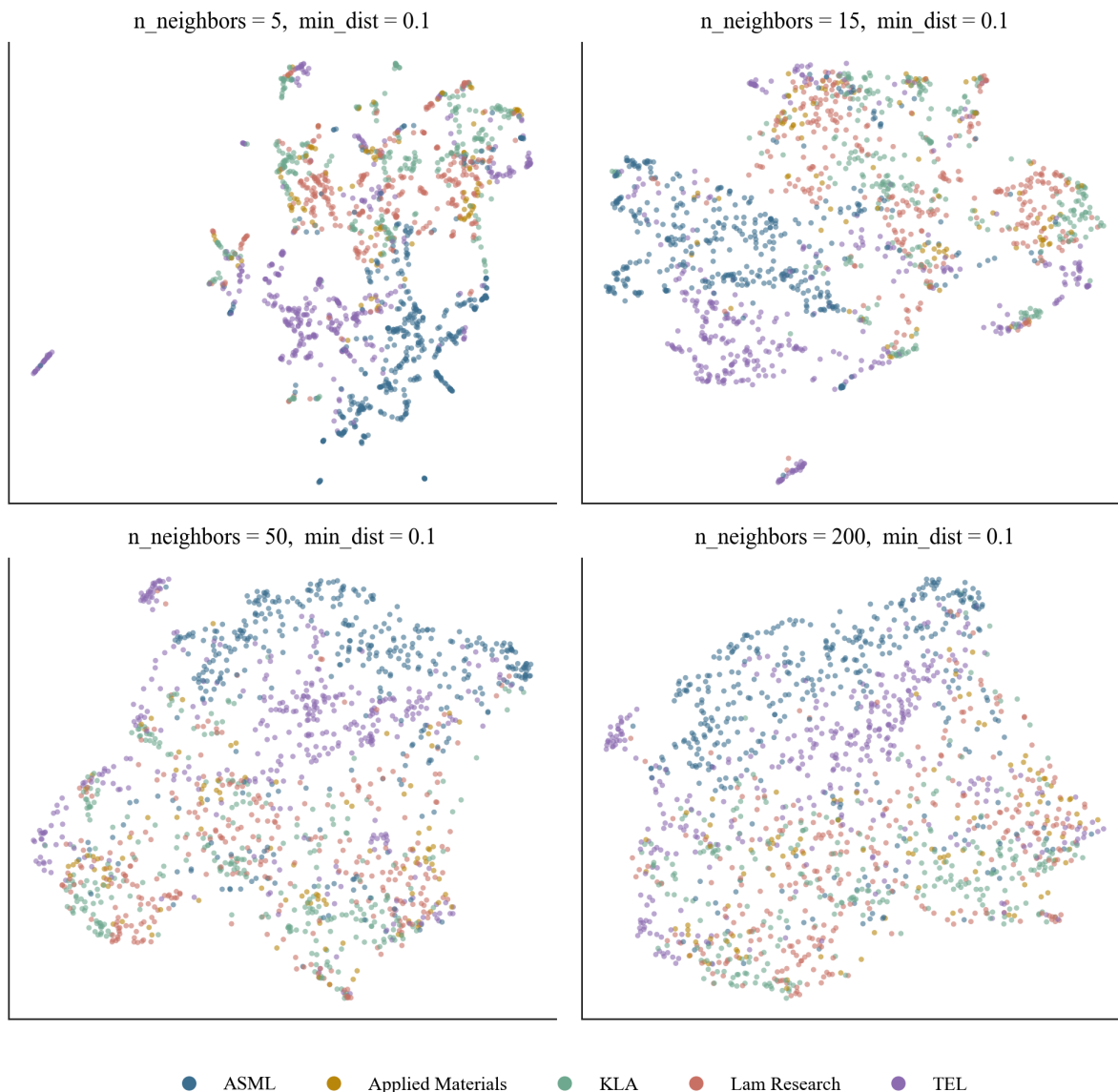


Figure 11: UMAP projections of the same corpus sample at four values of $n_neighbors$, with min_dist held fixed. Local vs. global emphasis is a user-facing choice; the bimodality is not.

8 Discussion

The finding is not that these five companies write differently from one another. They write very similarly; at the chunk level our two-dimensional projections do not resolve company identity. The finding is that any single report is itself already a bimodal object: it contains a narrative track and a tabular track, and the geometry of the embedding space picks up on that structure — as a displacement of class centroids — much more cleanly than it picks up on authorship. A retrieval system that treats every chunk as a member of a single population, and every top-k neighbourhood as a semantically coherent context, is likely to fail differently on the two tracks. The prose track will support “what does this company plan to do about X?” questions; the tabular track will pollute them with vocabulary-overlapping but intent-mismatched context.

The corresponding design implication is that if the two tracks are learnable from the raw embeddings — §2 shows they are — then a downstream system can route on them. One natural route is separate indices with separate top-k budgets; a lighter one is a single index re-ranked by the query’s projection onto the

class-centroid axis. Which is worth doing depends on the query mix a system is served, and on how often the two tracks are answering the same question in incompatible languages.

The present study is limited in three ways worth naming. First, only one industry is represented, so all industry-level claims remain hypotheses. Second, only one embedding model is used; the anisotropy figure would look different for a model trained with a stronger uniformity objective, and the class-centroid separation would potentially sharpen or blur. Third, the content-class label is regex-based and coarse; a trained classifier might refine it. None of these limitations undermines the diagnostic claim, which is elementary: look at the class centroids of your corpus in embedding space before you build on top of it.

9 Future directions

The measurements above are deliberately minimal — one sector, one encoder, one chunk size — and even at that minimum, class-centroid displacement is doing something interpretable and actionable. Five directions follow naturally.

9.1 Chunk-window ablation

Every result is computed on ~900-character chunks with a 120-character overlap. A systematic ablation across, say, {300, 900, 2000} characters would let us ask whether the class-centroid axis sharpens as chunks lengthen (more context per point) or blurs (mixing registers inside one window). Intuition points in opposite directions for the two ends: prose should benefit from more context, tables should become harder to identify as prose leaks in.

9.2 Cross-encoder comparison

MiniLM was chosen for compactness, not for state-of-the-art geometry. Encoders trained with explicit uniformity objectives (SimCSE) or with a contrastive-decoder architecture (Karpukhin et al., 2020) would occupy the sphere differently and would potentially reduce the anisotropy gap of Figure 7. The question is whether the class-centroid displacement survives the encoder swap; a grid of encoders on the same corpus turns the current single-encoder diagnostic into a comparative statement about representation quality.

9.3 Cross-industry expansion

The corpus here is one sector. Adding sectors whose reporting is likely to look qualitatively different — oil and gas, pharmaceuticals, consumer banking, apparel — would let us ask whether the financial↔narrative class-centroid axis is universal (a property of the disclosure genre) or sector-specific (a property of what each sector must quantify).

9.4 Interpretability of the class-centroid direction

The direction $\mu_{\text{tabular}} - \mu_{\text{narrative}}$ is itself a 384-dimensional vector, and its projection onto known token-level features (digit density, punctuation frequency, mean line length, presence of currency symbols) would decompose the axis into human-readable components. This is the same species of question mechanistic interpretability asks about individual attention heads, exported outward to the corpus level.

9.5 Centroid-shift analyses in other document classes

The claim of §2 is general. Any collection of text whose author-declared structure mixes distinguishable content classes into a single stream admits the same diagnostic: compute the class centroids, plot their displacement, and read the axis. Several domains present themselves immediately.

In scientific writing, the methods and discussion sections of a paper mix hypothesis and procedure; the centroid displacement across sections of the same paper is a candidate signal for measuring how tightly a field’s reporting norms hold. In clinical records, structured history and free-text notes describe the same

encounter in incompatible registers; the class-centroid axis would give a per-record score of how narratively vs. procedurally the encounter is being documented, with direct implications for downstream extraction. In legal filings, statutory-citation passages and argumentative passages have distinct rhetorical structure; their centroid displacement is a candidate axis for a document-triage tool for practitioners handling large brief volumes. In parliamentary or committee transcripts, speaker-A vs. speaker-B centroids trace who occupies which rhetorical register on which topic. In research grants, the technical vs. rhetorical passages of a proposal have measurable centroid separation, and the ratio of chunks on each side is a proxy for how specifically the proposal has been costed.

In each case the diagnostic is one line of code once the corpus is embedded: compute $\mu_i - \mu_j$, project every chunk onto that difference vector, and inspect. The value of the framing is not in any single instance but in the fact that a corpus reader now has an inexpensive, general-purpose first look at whether the structure the authors of a corpus claim to have imposed on it has left a geometric footprint the encoder can see.

9.6 Policy tooling

For the policy audience of §1.2, the most immediate artefact is a per-report score along the class-centroid axis of §2. Given a target set of disclosure topics (Scope 3 emissions, water withdrawal, supplier auditing coverage), one can ask how many chunks a report devotes to each topic and how many land on the numeric versus the narrative side of the axis. That statistic is a first-order proxy for disclosure quality — not a substitute for expert audit, but a useful triage tool over the 200-plus reports published each year by the S&P 500 alone.

References

- Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP. arXiv:1908.10084.
- Ethayarajh, K. (2019). How Contextual are Contextualized Word Representations? Comparing the Geometry of BERT, ELMo, and GPT-2 Embeddings. EMNLP.
- Li, B., Zhou, H., He, J., Wang, M., Yang, Y., and Li, L. (2020). On the Sentence Embeddings from Pre-trained Language Models. EMNLP. arXiv:2011.05864.
- Gao, T., Yao, X., and Chen, D. (2021). SimCSE: Simple Contrastive Learning of Sentence Embeddings. EMNLP. arXiv:2104.08821.
- van der Maaten, L. and Hinton, G. (2008). Visualizing Data using t-SNE. *Journal of Machine Learning Research* 9, 2579–2605.
- McInnes, L., Healy, J., and Melville, J. (2018). UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. arXiv:1802.03426.
- Coenen, A., Reif, E., Yuan, A., Kim, B., Pearce, A., Viégas, F., and Wattenberg, M. (2019). Visualizing and Measuring the Geometry of BERT. NeurIPS.
- Karpukhin, V., Oğuz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., and Yih, W.-t. (2020). Dense Passage Retrieval for Open-Domain Question Answering. EMNLP.
- Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS.
- Thakur, N., Reimers, N., Rücklé, A., Srivastava, A., and Gurevych, I. (2021). BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. NeurIPS Datasets and Benchmarks.
- Muennighoff, N., Tazi, N., Magne, L., and Reimers, N. (2023). MTEB: Massive Text Embedding Benchmark. EACL.
- Sarathi, P., Abdullah, S., Tuli, A., Khanna, S., Goldie, A., and Manning, C. D. (2024). RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval. ICLR.